

norming-spanish

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Exp info

- Experiment design

The participants will sign a consent form, go through a headphone screening test, and then move on to the norming experiment. In this experiment, they will go through 23 VOT steps x 4 repetitions/blocks x 3 word pairs + 12 catch trials = 288 trials in total. Each participant hears all VOT steps across all word pairs (only within-subject). On catch trials, participants are instructed to press either left or right.

The VOT steps are as follows: -100, -50, -20, -10, 0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 80, 90, 100, 120

- Randomization

The order of trials was randomized for each participant. Catch trials were distributed randomly throughout the experiment.

Participants were given the opportunity to take breaks after each block. Participants took about 20 minutes to complete all the trials. For each participant, /d/-initial words were either always displayed on the left side or always displayed on the right side. Across participants, this ordering was counter-balanced.

Preprocessing and exclusion

- Exclusion standards:

Participants who committed more than 3 errors out of the 12 catch trials will be excluded ($N = 2$).

Participants with an average RT more than three standard deviations from the mean of the by-participant means ($N = 0$) will be excluded.

Trials that are three SDs from the participant's mean RT were excluded from analysis ($n=80$) will be excluded.

Plotting

General principles for wrangling and plotting

1. start from one data point
2. start from raw data
3. always include uncertainty

```
# prepare the data for plot
data.plot <- data.preprc %>%
  select(VOT, resp.num, ID, block, word, stim, rt, rt.log)

summary(data.plot)
```

```
##           VOT           resp.num           ID           block
##  Min.      :-100.00  Min.      :0.0000  Min.      :15404649  Min.      :1.00
##  1st Qu.:   5.00   1st Qu.:0.0000   1st Qu.:15432564   1st Qu.:1.75
##  Median :  35.00   Median :0.0000   Median :15432582   Median :2.50
##  Mean   :  31.96   Mean   :0.4489   Mean   :15430654   Mean   :2.50
##  3rd Qu.:  65.00   3rd Qu.:1.0000   3rd Qu.:15432596   3rd Qu.:3.25
##  Max.    : 120.00   Max.    :1.0000   Max.    :15432784   Max.    :4.00
##           word           stim           rt           rt.log
##  Length:7728   Length:7728   Min.      : 7.7   Min.      :0.8865
##  Class :character  Class :character  1st Qu.: 331.1   1st Qu.:2.5199
##  Mode  :character  Mode  :character  Median : 502.3   Median :2.7010
##                                     Mean   : 655.4   Mean   :2.7045
##                                     3rd Qu.: 774.4   3rd Qu.:2.8890
##                                     Max.    :5005.2   Max.    :3.6994
```

```
glimpse(data.plot)
```

```
## Rows: 7,728
## Columns: 8
## Groups: ID [28]
## $ VOT      <dbl> -10, -10, 80, 5, 35, 15, -100, 120, 65, 20, 100, 25, 40, 30, ~
## $ resp.num <dbl> 1, 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 1, 0, 1, 1, 0, 1, 1~
## $ ID       <int> 15406260, 15406260, 15406260, 15406260, 15406260, 15406260, 1~
## $ block    <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ word     <chr> "dill", "dip", "din", "dip", "dill", "din", "din", "din", "di~
## $ stim     <chr> "dill_-10.wav", "dip_-10.wav", "din_+80.wav", "dip_+5.wav", "~
## $ rt       <dbl> 1064.9, 1296.6, 510.6, 953.3, 627.1, 439.9, 328.4, 391.0, 496~
## $ rt.log   <dbl> 3.027309, 3.112806, 2.708081, 2.979230, 2.797337, 2.643354, 2~
```

Start from single subject data

As we are concerned about the difference of categorization functions of three contrasts, I made “word” a grouping variable for every plot.

Through data from the single subject, we can already see that the behavior of “din” contrast is different from that of “dip” and “dill”: the number of 0, 1 response are closer to each other; (2) its overall mean is closer to 0.5, (3) its categorization curve shifts earlier

```

# choose 1 subject and summarise mean_resp
data.plot %>%
  filter(ID == '15432596') %>%
  group_by(word, VOT) %>%
  summarize(mean_resp.word = mean(resp.num),)

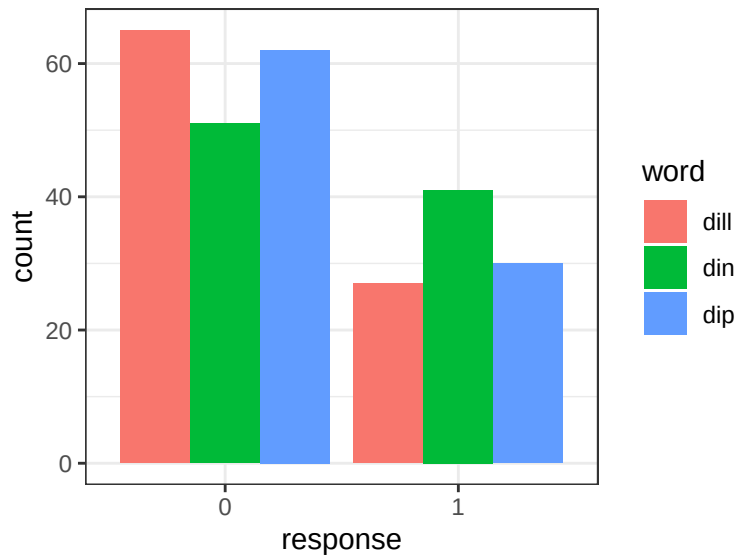
## 'summarise()' has regrouped the output.
## i Summaries were computed grouped by word and VOT.
## i Output is grouped by word.
## i Use 'summarise(.groups = "drop_last")' to silence this message.
## i Use 'summarise(.by = c(word, VOT))' for per-operation grouping
## ('?dplyr::dplyr_by') instead.

## # A tibble: 69 x 3
## # Groups:   word [3]
##   word    VOT mean_resp.word
##   <chr> <dbl>         <dbl>
## 1 dill  -100             0
## 2 dill  -50             0
## 3 dill  -20             0
## 4 dill  -10             0
## 5 dill   0             0
## 6 dill   5             0
## 7 dill  10             0
## 8 dill  15             0
## 9 dill  20             0
## 10 dill  25             0
## # i 59 more rows

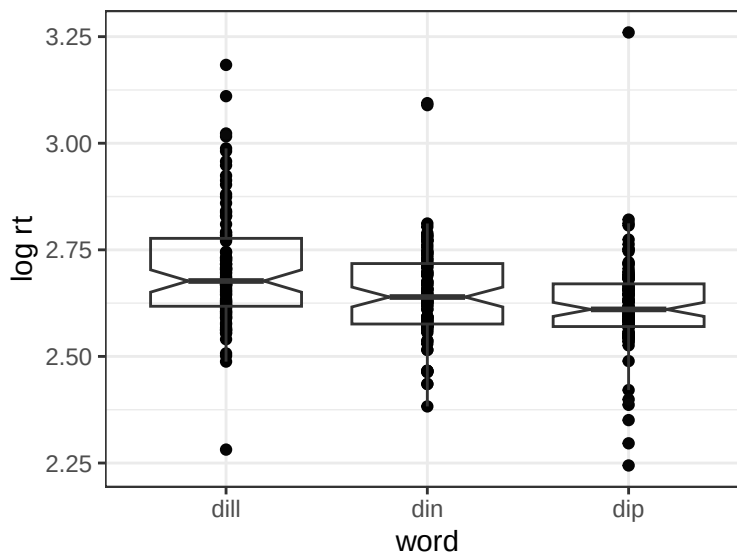
data.s1 <- data.plot %>%
  filter(ID == '15432596')

# simple count the number of 0, 1 responses for each word
p.s1.count <- ggplot(data.s1, aes(x = factor(resp.num), fill = word)) +
  geom_bar(position = "dodge") +
  labs(x = "response", y = "count")
p.s1.count # d is 0

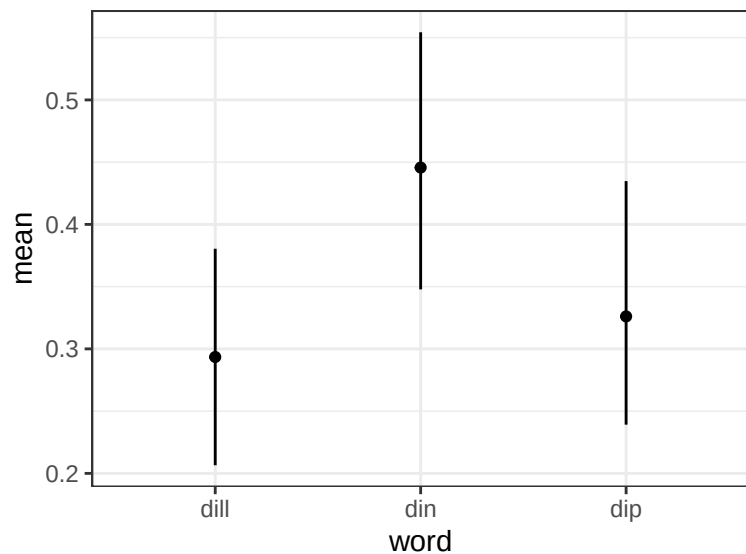
```



```
# take a look at rt just in case
p.s1.rt <- ggplot(data.s1, aes(x = factor(word), y = rt.log)) +
  geom_point()+
  geom_boxplot(fill = NA, outlier.alpha = .1, notch = T)+
  labs(x = "word", y = "log rt")
p.s1.rt # d is 0.
```

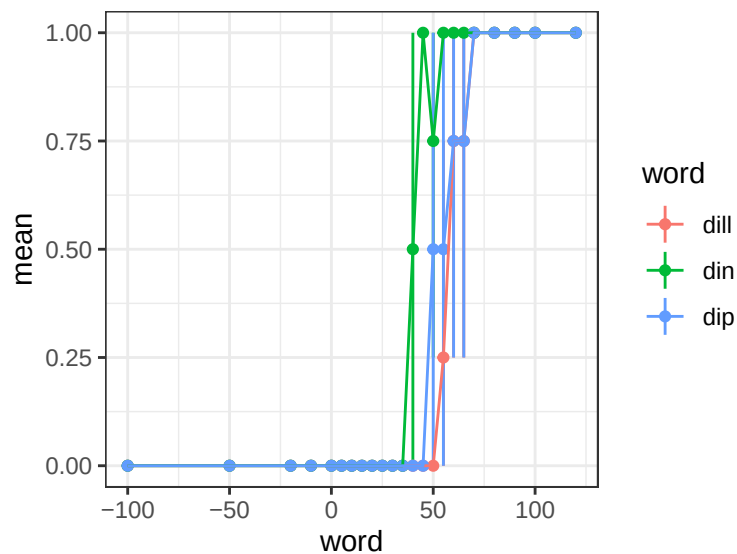


```
# per subj, per word --> 69 data points each
p.s1.resp <- ggplot(data.s1, aes(x = factor(word), y = resp.num)) +
  stat_summary(fun.data = mean_cl_boot, geom = "pointrange", size = .2) +
  labs(x = "word", y = "mean")
p.s1.resp
```



```
# per subj, per vot, per word -> 4 data points each
p.s1.vot <- ggplot(data.s1, aes(x = VOT, y = resp.num, color = word)) +
  stat_summary(fun.data = mean_cl_boot, geom = "pointrange", size = .2) +
  stat_summary(fun.data = mean_cl_boot, geom = "line") +
  # geom_smooth(method = "glm", method.args = list(family = "binomial")) #avoid making the plot too crowdy

# # ver2, no uncertainty around mean
# stat_summary(fun = mean, geom = "point") +
# stat_summary(fun = mean, geom = "line") +
# geom_smooth(method = "glm", method.args = list(family = "binomial"))
labs(x = "word", y = "mean")
p.s1.vot
```



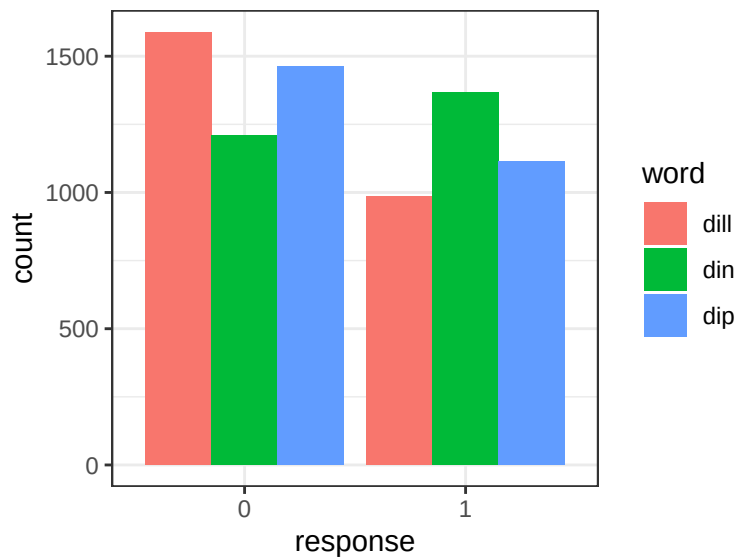
move to multiple subj

Verify preliminary findings in multiple subjects.

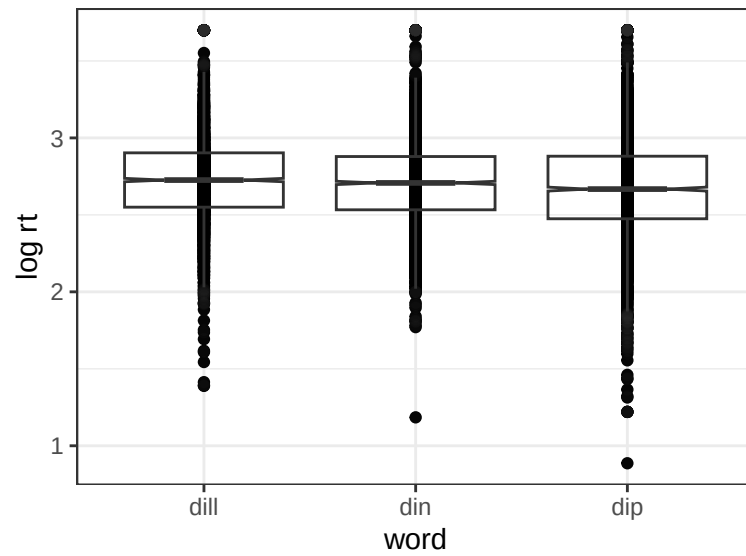
```
p.count = p.s1.count %+\n  data.plot
```

```
## Warning: <ggplot> %+\n  x was deprecated in ggplot2 4.0.0.\n## i Please use <ggplot> + x instead.\n## This warning is displayed once per session.\n## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was\n## generated.
```

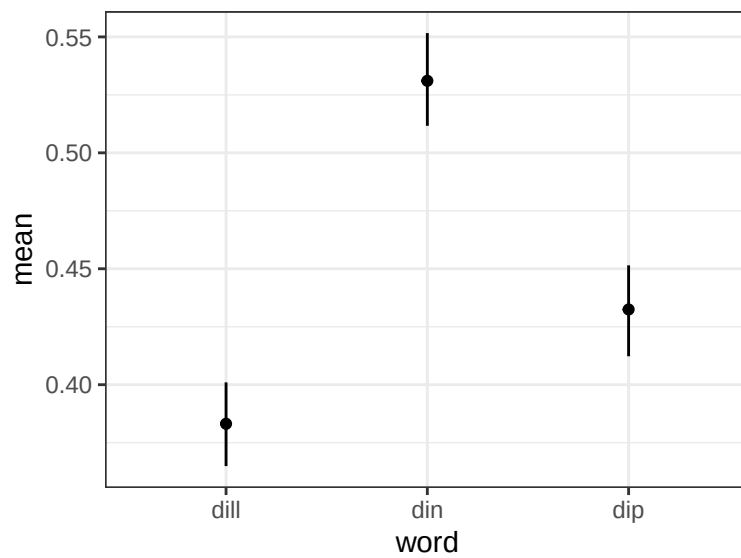
```
p.count
```



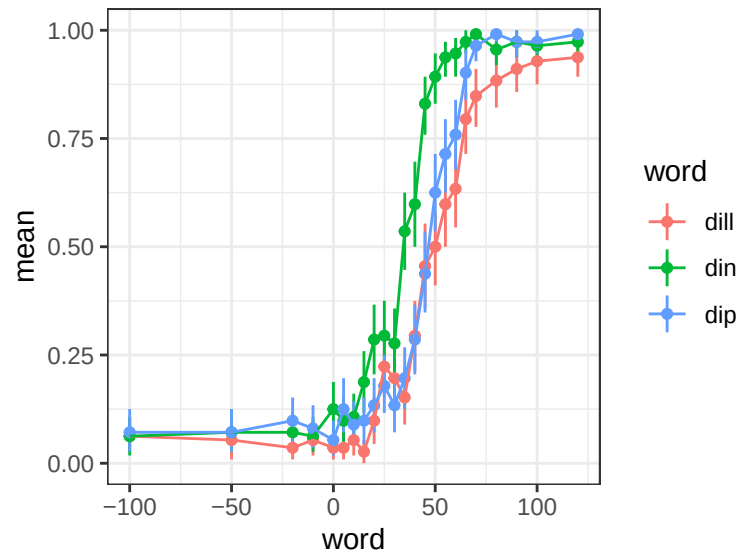
```
p.rt = p.s1.rt %+\n  data.plot\np.rt
```



```
p.resp = p.s1.resp %>%
  data.plot
p.resp
```

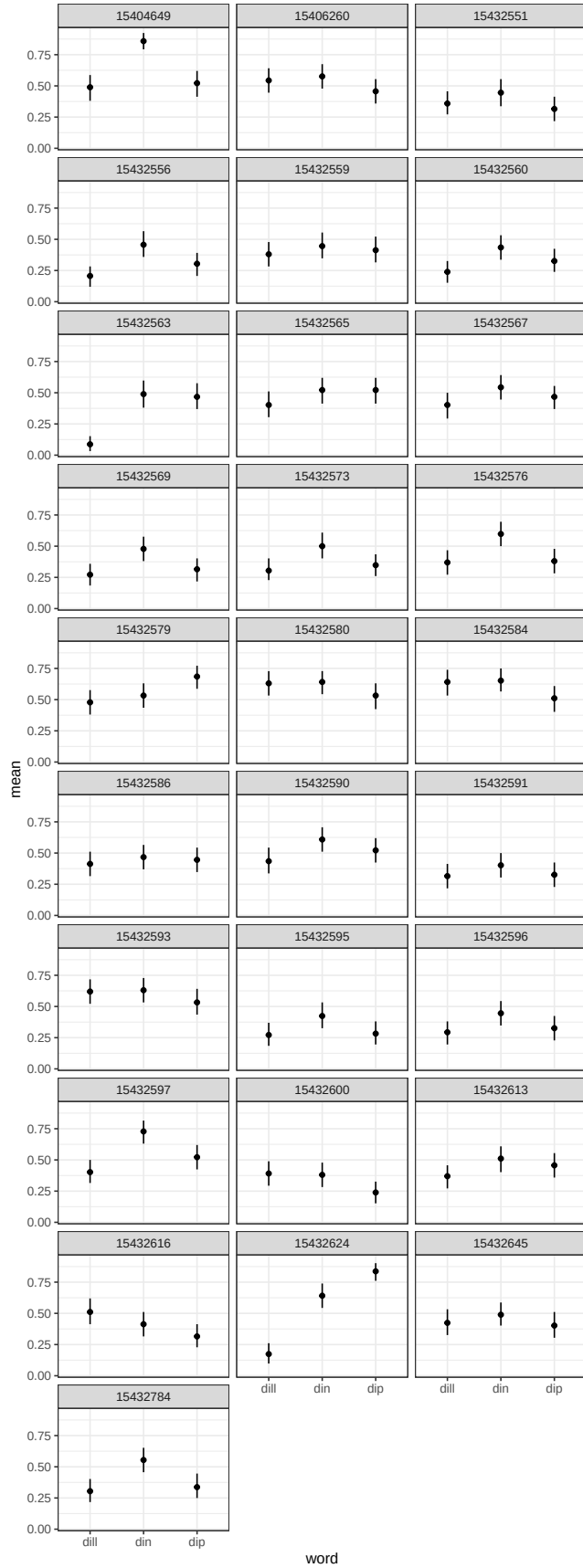


```
p.vot = p.s1.vot %>%
  data.plot
p.vot
```



Check subject-by-subject data.

```
p.resp+
  facet_wrap(facets = ~ ID, ncol =3)
```



Analysis

Our goal is to evaluate whether categorization functions for the three minimal pairs differs a lot from each other. And one minimal pair indeed differs.

```
data.preprc %<>%
  mutate(VOT.z = as.numeric(scale(VOT)),
         word = factor(word))

model <- glmer(resp.num ~ VOT.z * word + (1 + VOT.z | ID),
  data = data.preprc,
  family = binomial)

summary(model)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##   Family: binomial   ( logit )
## Formula: resp.num ~ VOT.z * word + (1 + VOT.z | ID)
##   Data: data.preprc
##
##           AIC          BIC      logLik -2*log(L)  df.resid
##      4763.1      4825.7   -2372.5    4745.1      7719
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -119.46   -0.29    0.00    0.28   797.88
##
## Random effects:
##   Groups Name            Variance Std.Dev. Corr
##   ID      (Intercept)  1.926      1.388
##           VOT.z        5.927      2.435   -0.71
## Number of obs: 7728, groups: ID, 28
##
## Fixed effects:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.77343    0.27567  -6.433 1.25e-10 ***
## VOT.z         4.59541    0.48815   9.414 < 2e-16 ***
## wordddin      1.47343    0.09690  15.206 < 2e-16 ***
## wordddip      0.44492    0.09431   4.718 2.39e-06 ***
## VOT.z:wordddin 0.64861    0.16613   3.904 9.45e-05 ***
## VOT.z:wordddip 0.25568    0.15431   1.657 0.0975 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) VOT.z  wordddn worddp VOT.z:wrddn
## VOT.z         -0.694
## wordddin      -0.201  0.104
## wordddip      -0.184  0.075  0.514
## VOT.z:wrddn   0.048 -0.128 -0.143 -0.181
## VOT.z:wrddp   0.063 -0.147 -0.186 -0.340  0.447
```